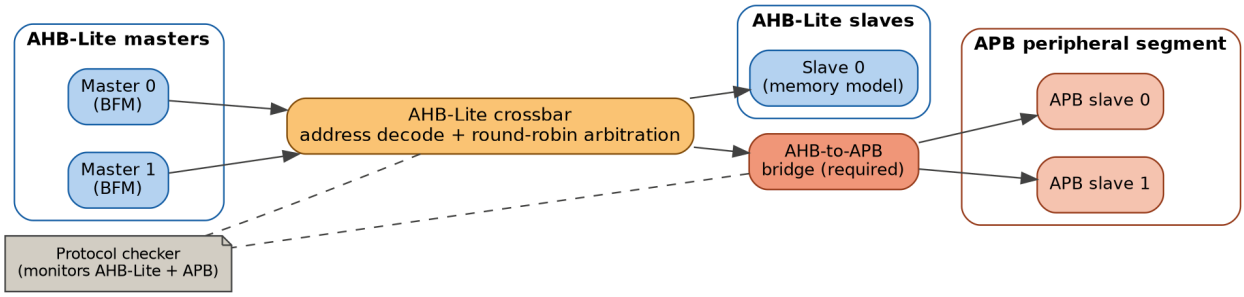
AMBA AHB-APB Capstone Project

AMBA (Advanced Microcontroller Bus Architecture) is one of the most popular interface standards for developing system-on-chip (SoC) devices. This project will involve developing semiconductor IP cores and verification IP for the AHB-Lite protocol. Specifications from Arm can be downloaded from here.



The above diagram shows the components of the system:

- AHB-Lite master bus-functional model (verification software)
- AHB-Lite slave bus functional model (verification software)
- Protocol checker (verification software)
- AHB-Lite crossbar (hardware core)

The entire design will be done in the Chisel/Scala language in accordance with the chiselWare Standard which can be downloaded from chiselWare Developers Kit on GitHub. The Developer's kit contains the complete software development in containers, including support for VS Code Dev Containers.

Requirements

Required Scope

The following are the required features of the cores.

- APB master BFM and APB slave BFM (parameterized register file / memory model with configurable wait-state injection via PREADY).
- AHB-Lite master BFM (transaction-item driven) and AHB-Lite slave BFM (parameterized memory model, configurable HREADYOUT wait states).
- AHB-Lite protocol checker: validates address-phase/data-phase pipelining, HTRANS legality, and correct HREADYOUT handshaking.
- APB protocol checker: validates the setup/access/enable state sequence and PREADY behavior.
- N x M AHB-Lite crossbar with static address-map decode and round-robin arbitration across masters (parameterize N and M; must support at least 2 masters x 2 slaves for the graded demo).
- AHB-to-APB bridge, integrating the APB bus as a peripheral segment reachable through the AHB-Lite crossbar's address map.
- Directed and constrained-random test suites (per Section 7) for every block above, including the bridge.
- FPGA demo: a 2-master x 2-slave AHB-Lite crossbar instance, with at least one AHB-Lite slave reached directly and at least one APB peripheral reached through the bridge, driving representative traffic with pass/fail shown on-board.

Stretch goals

The following are option features that the team can add as time permits.

- Full AHB5 feature set: additional burst types (WRAP), protection signaling (HPROT), and/or exclusive access.
- Multi-layer AHB interconnect supporting simultaneous transactions to different slaves from different masters (true crossbar fan-out rather than single shared bus).
- Expanded crossbar size (e.g., 8x8) with a configurable/priority-based arbiter as an alternative to round-robin.
- A small coverage-driven (functional coverage) verification environment layered on top of the constrained-random tests.
- Multiple APB peripheral segments behind independent bridges, exercising address-map partitioning across more than one bridge instance.

ChiselWare Packaging Requirements

chiselWare has very specific requirements for what a chiselWare core looks like. These are codified in the chiselWare Standard. Detailed information on designing to this specification is

found in the chiselWare Developers Guide. Both can be found in the Developers Kit found on GitHub as previously mentioned.

The chiselWare standard however has not yet addressed the standardization of bus functional models (BFM). The team will therefore develop their own approach in collaboration with the sponsor with an aim that their work will one day make its way into the chiselWare Standard.

Verification Requirements

The chiselWare standard calls out standards for verification and those will apply to this project as well.

- Unit-level tests for every bundle/module using chiseltest, covering both directed and constrained-random stimulus.
- The AHB-Lite and APB protocol checkers
- A reusable transaction-item abstraction for BFMs so the same driver supports directed tests, randomized regression.
- Crossbar-specific scenarios: back-to-back transactions to the same slave, contention across multiple masters, and boundary/error cases (e.g., decode to an unmapped address).
- Bridge-specific scenarios: transactions that cross from AHB-Lite into APB and back, including wait-state injection on the APB side and its effect on AHB-Lite pipelining.